

Adaptive Learning Challenges in Autism: Reduced Evidence Accumulation and Prior Belief Updating in Probabilistic Reversal Reward Learning



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Background

Real-World Challenge

Children constantly adapt to dynamic, socially rich environments, a process that is particularly demanding for autistic children.

What Do We Know?

- Autistic children show deficits in **adaptive behavior**, struggling to flexibly respond to changing environments.
- They also demonstrate difficulties in **reward-based learning**, failing to efficiently process social feedback.
- Yet how these two challenges **interact at the cognitive level** remains unclear.

This study uses two powerful tools:

Framework	Role
Probabilistic Reversal Reward Learning (PRRL)	Captures how children update choices when reward contingencies shift
Drift Diffusion Model (DDM)	Decomposes decision-making into interpretable cognitive parameters

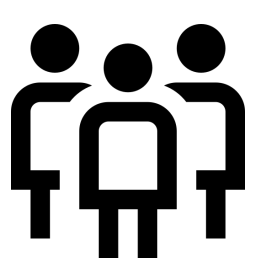
Objectives

To compare autistic vs. typically developing (TD) children on:

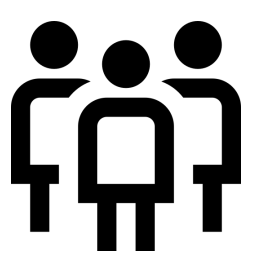
- Behavioral performance in PRRL
- Underlying decision-making mechanisms with computational model

Methods

Participants



ASD group (n = 35)
♂ : ♀ = 31 : 4
Age: 9.15 ± 2.09 years



TD group (n = 41)
♂ : ♀ = 21 : 20
Age: 9.15 ± 1.64 years

Task Design

Stimuli

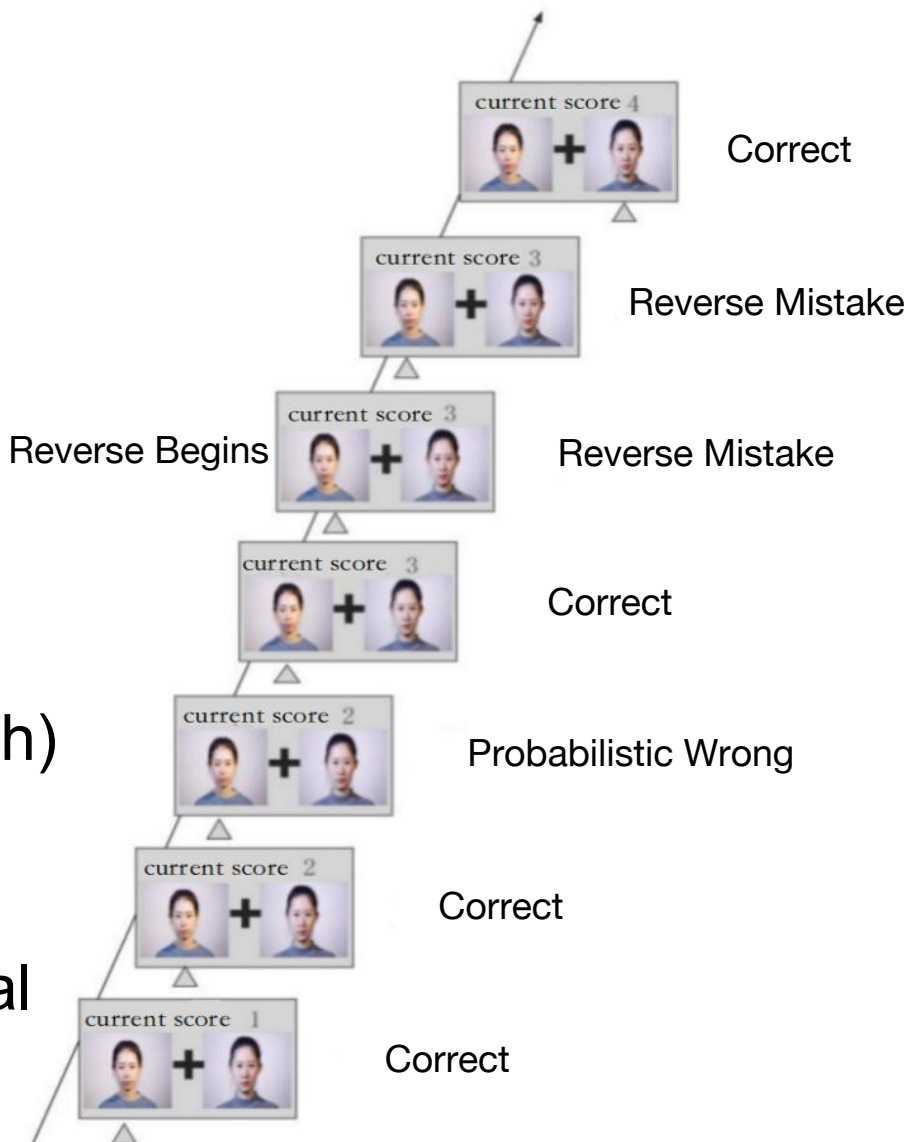
- Two neutral faces
- A: 70% reward ✓ B: 30% reward ⚠

Reversal

- Contingencies switch at Trial 30
- Total: 60 trials divided into 6 stages (10 trials each)

Goal

Maximize nonsocial rewards by selecting the optimal face



Behavioral & Computational Indices

Index	Definition
Advantageous choice rate	Overall accuracy across all 60 trials
Switching cost	Stage 4 – Stage 3 (reversal cost)
Recovery rate	Stage 5 – Stage 4 (post-reversal adaptation)
Drift rate (DDM)	Evidence accumulation efficiency
Decision boundary (DDM)	Response threshold

Results

Behavioral Performance

- TD children showed higher advantageous choice rates than ASD children ($F(1,72) = 20.10, p < .001$)
- ASD children exhibited:
 - Lower switching costs** ($t(74) = 2.11, p = .038$)
 - Slower recovery after reversal** ($t(74) = 4.20, p < .001$)

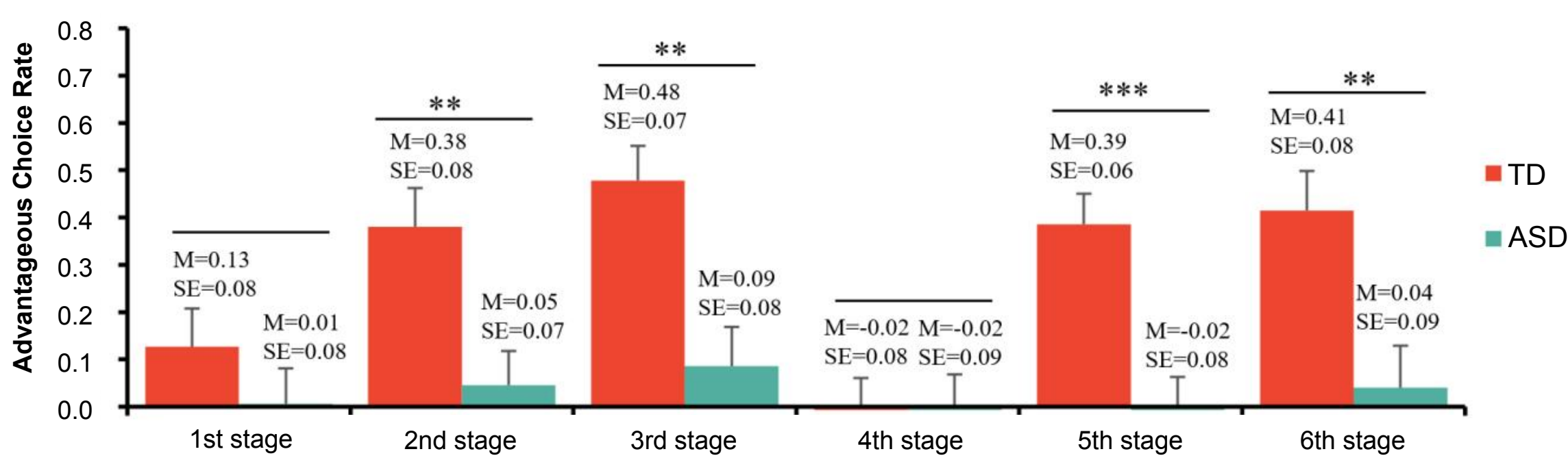


Figure 1. The six-stage average advantageous choice rate under the condition of social stimuli. The error bar is the standard error (SE); the task has a total of 60 trials, and 10 trials are divided into one stage. The fourth stage is the reversal stage. *** $p < .001$, ** $p < .01$.

Decision-Making Mechanism (DDM)

- ASD children showed a **lower drift rate** ($t(74) = 3.03, p = .003$)
- No significant group difference in decision boundary

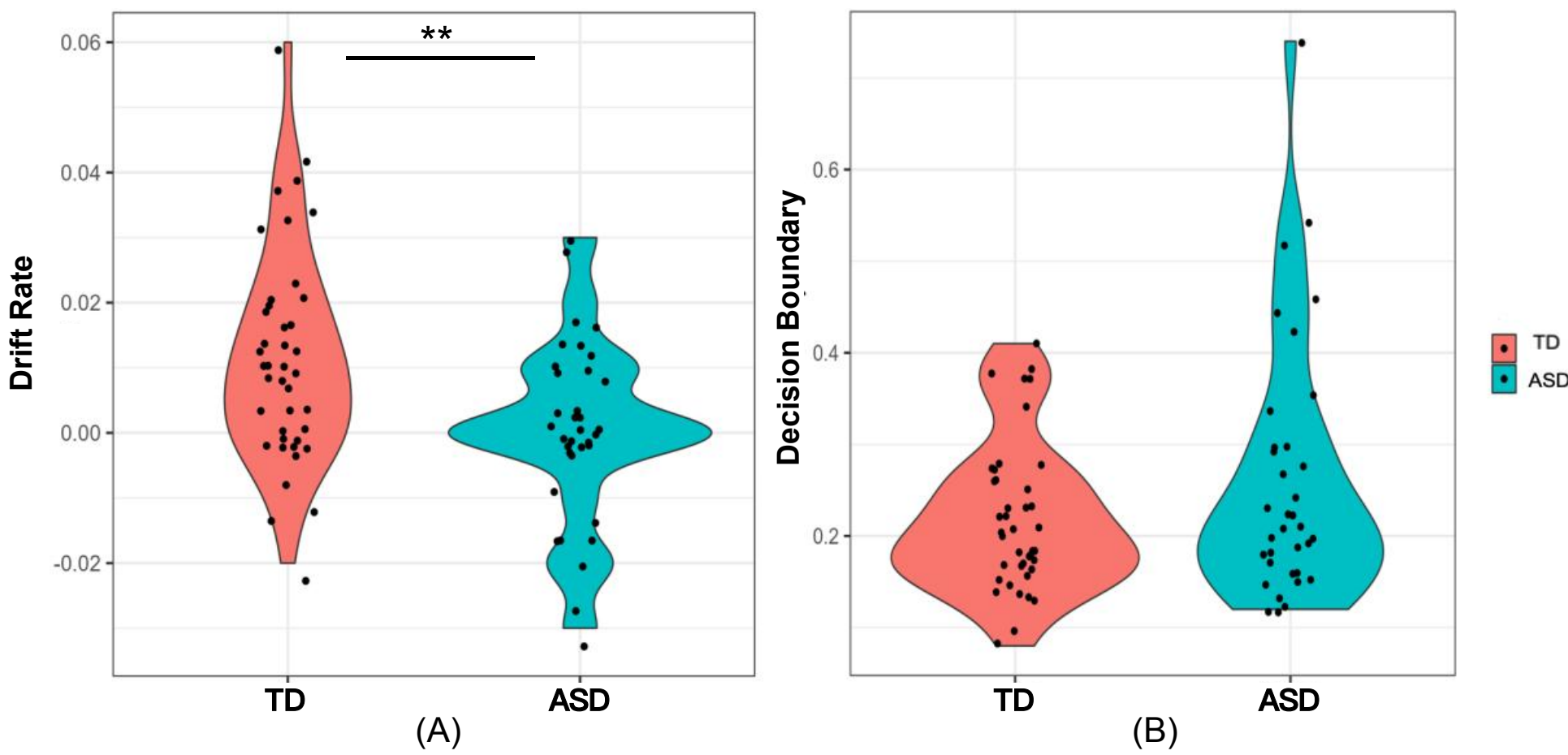


Figure 2. (A) Drift rate (v) and (B) decision boundary (α) in the probability reversal reward learning task.

Symptom Correlations

- Higher **autistic symptoms** were associated with **lower drift rates** and **slower recovery rates**

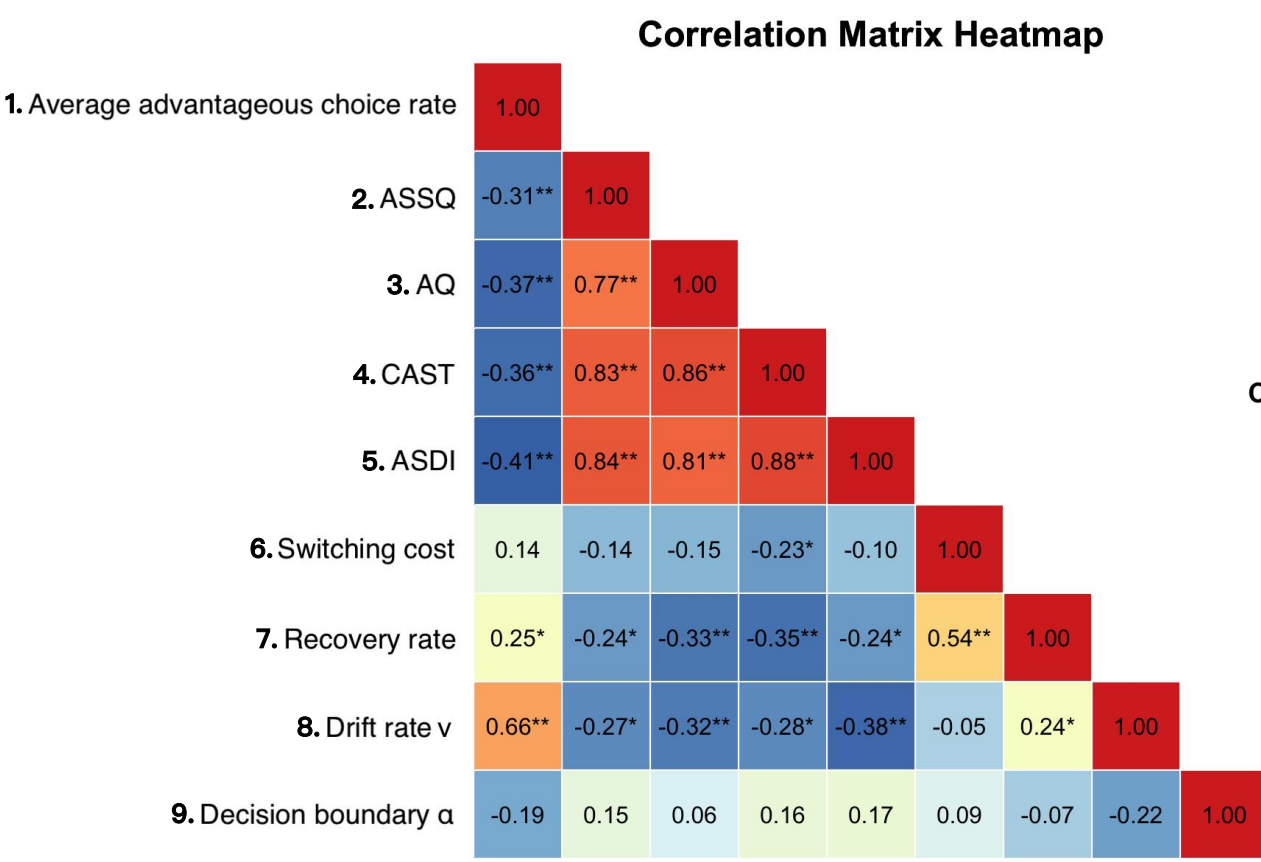


Figure 3. Heat map to indicate the Pearson correlations between average advantageous choice rate, autistic symptoms, conversion cost, recovery rate, and EZDDM model parameters (drift rate v and decision boundary α). ASSQ: the Autism Spectrum Screening Questionnaire score; AQ: the Autism Quotient score; CAST: the Childhood Autism Spectrum Test score; ASDI: the Asperger Syndrome Diagnostic Interview Inventory score. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Conclusion

Key Findings

- Autistic children show **reduced reliance on prior reward associations** and **slower evidence accumulation** during reversal learning.
- These mechanisms together contribute to **poorer adaptability** in volatile, social environments.

Clinical Implications

Findings support the development of targeted interventions, such as:

- Patient, **long-term training** to strengthen reward-based learning
- Context-dependent approaches** that explicitly support the formation and updating of prior beliefs about social and environmental contingencies.